Supplementary Material

Duplicate Data Rows Make Prediction Uncertainty Look Better Than It Is: A Simple Check and Fix for New Cases

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S1 Proofs

Throughout, Assumption A1 holds: the rows are split uniformly at random, so given the set of training rows the other rows are assigned to calibration and test at random, and the seen/new label κ_T is a fixed function of the training features.

Equation (2). By the law of total probability, $c = \pi c_{\mathrm{S}} + (1-\pi)c_{\mathrm{N}}$; solving for c_{N} gives the result. Split conformal prediction gives $1-\alpha \le c \le 1-\alpha+1/(n_C+1)$ when scores have no ties (as in [1]). Hence, with $m_{\mathrm{S}} = 1-c_{\mathrm{S}}$, the miss rate on new cases $m_{\mathrm{N}} = 1-c_{\mathrm{N}}$ satisfies

$$\frac{\alpha - 1/(n_C+1) - \pi m_{\mathrm{S}}}{1-\pi} \le m_{\mathrm{N}} \le \frac{\alpha - \pi m_{\mathrm{S}}}{1-\pi}. \tag{S1}$$

Exact duplicates in the calibration set create tied scores; ties keep the lower bound on coverage but can break the upper one.

Equation (3) (standard combinatorics). Given that row j is a test row, the training set is a uniformly random n_T-subset of the other $N-1$ rows. Row j is new exactly when none of its $m_j - 1$ copies is chosen, which has probability $\binom{N-m_j}{n_T}/\binom{N-1}{n_T}$. Averaging over rows gives Eq. (3).

Proposition 1 (closed-form label-pure exposure). Let row j have $m_j - 1$ copies, d_j of which carry a different label. Row j is label-pure seen if no differently labelled copy is chosen for training but at least one copy is. The event "no differently labelled copy chosen" has probability $\binom{N-1-d_j}{n_T}/\binom{N-1}{n_T}$ and contains the event "no copy chosen", of probability $\binom{N-m_j}{n_T}/\binom{N-1}{n_T}$. The difference is the probability of the event of interest; averaging over rows gives Eq. (4). *Check:* the closed form agreed with an average over 30 random splits to within 0.005 on five data sets (kc1, pc1, analcatdata_supreme, black_friday, wine_quality), and the tool's unit test confirms it by simulation. □

Proposition 2 (memorization limit). Call a row memorizable if its training copies all carry exactly its label, and let π_P be the probability that a calibration or test row is memorizable. Under Assumptions A2 and A3, calibration scores are draws from the mixture $\pi_P\delta_0 + (1-\pi_P)F$ with F continuous. The conformal threshold is an empirical quantile at level $\lceil (n_C+1)(1-\alpha)\rceil / n_C \to 1-\alpha$; by the law of large numbers for empirical quantiles it converges to the $(1-\alpha)$-quantile of the mixture. If $\pi_P < 1-\alpha$, this is the point t with $\pi_P + (1-\pi_P)F(t) = 1-\alpha$, so a non-memorizable row (in particular a new row) is covered with probability $F(t) = 1 - \alpha/(1-\pi_P)$, a gap of $-\alpha\pi_P/(1-\pi_P)$. If $\pi_P \geq 1-\alpha$ the threshold is 0 and non-memorizable rows are covered with probability $F(0) = 0$. In a simulation with $\pi_P = 0.4$ and 1,000 calibration rows (800 repetitions), the coverage of non-memorizable rows was within 0.01 of $1 - 0.1/0.6$ (unit test `test_purity_limit_simulation`). □

Proposition 3 (validity of separate calibration; standard). Let $\hat{q}_k$ be the $\lceil (n_k+1)(1-\alpha)\rceil$-th smallest score among the n_k calibration rows of group k ($+\infty$ if that rank exceeds n_k). Condition on T, on the group of test row j and on the multiset of group-k rows in $C \cup \{j\}$. Because κ_T is fixed given T and the remaining rows are exchangeable, each of these $n_k + 1$ rows is equally likely to be the test row, so the rank of the test score among them is uniform and $\Pr\{Y_j \in \mathcal{C}(X_j) \mid \kappa_T(X_j) = k\} \geq 1-\alpha$ (the Mondrian argument of [3]). Any deployment mix of seen and new cases is a weighted average of two quantities that are each at least $1-\alpha$. □

Safety margin (standard). With continuous scores, the coverage of the threshold at rank r among n calibration scores has the $\mathrm{Beta}(r, n-r+1)$ distribution over calibration draws. Choosing the smallest r with $\Pr\{\mathrm{Beta}(r, n-r+1) < 1-\alpha\} \leq \delta$ gives coverage at least $1-\alpha$ with probability at least $1-\delta$ [2].

S2 Hypothesis register

Every hypothesis of Table 3 was fixed in a version-control commit before its data were analysed: H1–H4 in the benchmark protocol (commit `0e58bdd`, 2026-10-20), H5, H6, H8 and H9 in the revised protocol (`91ac5f6`, 2026-10-21), and the wave-2 tests including H7 in Amendment 2 (`b5bbd25`, 2026-11-02, before any wave-2 data were downloaded). The register, with metrics, tests and outcomes, is in `research_design/experimental_protocol.md`. Analyses added after results were seen are labelled *post hoc* in the main text: the exposure strata and decomposition of the benchmark results, the retrospective test of Eq. (5) on the benchmark data, the sampling-noise check, the rule-based examples and class-conditional analysis, the planted-leak test, the cost measurement and the tool validation.

S3 Deviations from the plans

ID	Deviation (reason; effect)
D1	Output directories created by two experiment functions (bug found in a quick run; no effect).

ID	Deviation (reason; effect)
D2	Exploratory analyses added after the benchmark results: exposure strata, decomposition by Eq. (2), counts of gaps below −2 and −5 pp (to explain where the gap arises; labelled post hoc).
D3	Verification run on solar_flare with 200 extra splits (an apparent violation of Proposition 3 with 10 splits; confirmed as sampling noise: 90.9%, CI 89.9–91.8%).
D4	External application data sets (NASA mc1, pc2, kc3, mw1; KDD Cup'99) specified before their outcomes; KDD uniformly subsampled to 30,000 rows.
D5	A pre-registered cost experiment was run late (omission; no effect).
D6	Normalized-residual baseline added before its outcomes (anticipated reviewer request).
D7	KDD Cup'99 group split: infeasible splits (one group larger than a partition) had not been reported; now logged, and group-split results there use feasible splits only.
D8	Contingency amendment for an OpenML outage (not triggered).
D9	Wave 1: parser fix for list-valued OpenML metadata, before any data set was evaluated.
D10	Wave 1: four data sets listed twice in the pool; one copy kept (lowest id), before outcomes (24 to 20 data sets).
D11	Wave 1: five downloads failed for server-side reasons; retried for 60 minutes (cut-off fixed in advance); three stayed excluded.
D12	Wave 1 runs started early for the data sets fixed by the label-free rule (same code and configuration).
D13	Amendment 2: wave 2 (UCI), tie-breaking test and remedy comparison, pre-specified.
D14	Wave 2: unique column names for one file; resumed interrupted run; a fallback for one data set whose text column is entirely missing in a subset (used only where the standard path fails), all before wave-2 outcomes.
D15	Post hoc additions for the final paper: sampling-noise check, rule-based examples, planted-leak test, cost and memory, tool validation on fresh files.
D16	An earlier explanation of over-coverage by tied scores was refuted by the pre-specified tie-breaking test and replaced (no numbers changed).
D17	Manuscript moved to the journal's LaTeX template with all numbers generated from result files.

No data set, seed or learner was excluded after its outcome was seen, and no hyperparameter was tuned on test data.

S4 Selection of the fresh collections

S5 Per-data-set results on the fresh collections

Table S3: Wave 1 (AutoML Benchmark and Grinsztajn et al.), random forest, 20 splits. $\bar{\pi}$: expected share of seen test rows (%); g^*: predicted gap (pp); flag: combined rule. New-case coverage (%) for standard calibration (Std), separate calibration (Fix), with safety margin (Margin) and by distance (Dist.); – where the method could not be applied

Data set	Task	$\bar{\pi}$	g^*	Flag	Std	Fix	Margin	Dist.
analcatdata_supreme	regre.	93	−42.4	✓	62.5	89.3	94.3	92.8
airlines	class.	71	−5.5	✓	98.7	89.9	90.8	90.0
sf-police-incidents	class.	31	−3.2	✓	90.1	90.0	90.6	90.0
wine-quality-white	class.	23	−3.1	✓	87.5	90.4	91.9	90.5
wine_quality	regre.	22	−2.8	✓	87.6	90.2	91.3	90.5
compas-two-years	class.	65	−2.6	✓	92.4	89.4	91.3	90.0

Table S3: Wave 1 (AutoML Benchmark and Grinsztajn et al.), random forest, 20 splits. $\bar{\pi}$: expected share of seen test rows (%); g^*: predicted gap (pp); flag: combined rule. New-case coverage (%) for standard calibration (Std), separate calibration (Fix), with safety margin (Margin) and by distance (Dist.); – where the method could not be applied

Data set	Task	$\bar{\pi}$	g^*	Flag	Std	Fix	Margin	Dist.
tecator	regre.	10	−1.2	✓	91.3	91.3	96.1	96.3
Diabetes130US	class.	21	−1.0	✓	94.2	90.0	90.6	90.0
yeast	class.	3	−0.3		89.8	90.0	91.7	90.3
nyc-taxi-green-dec-2016	regre.	1	−0.1		90.0	90.1	90.6	90.2
MagicTelescope	class.	1	−0.1		89.8	89.9	90.6	90.1
superconduct	regre.	33	0.0	✓	89.1	89.9	90.6	89.9
Buzzinsocialmedia_Twitter	regre.	2	0.0		89.9	90.1	90.6	90.1
Bike_Sharing_Demand	regre.	3	0.0		90.3	90.3	91.0	90.3
seattlecrime6	regre.	100	0.0	✓	96.7	100.0	100.0	–
black_friday	regre.	63	0.0	✓	91.8	89.9	90.8	90.0
heloc	class.	6	0.0		89.9	90.5	91.4	90.6
particulate-matter-ukair-2017	regre.	2	0.0		90.0	89.9	90.4	90.0
Mercedes_Benz_Greener_Manufacturing	regre.	9	0.0		90.6	90.2	91.4	90.5
yprop_4_1	regre.	2	0.0		89.9	89.8	90.7	90.0

Table S4: Wave 2 (UCI repository), random forest, 20 splits. Columns as in Table S3

Data set	Task	$\bar{\pi}$	g^*	Flag	Std	Fix	Margin	Dist.
Skin Segmentation	class.	83	−49.2	✓	62.3	90.8	91.7	91.0
NATICUSdroid (Android Permissions)	class.	79	−15.8	✓	76.1	89.6	90.7	89.7
Early Stage Diabetes Risk Predic...	class.	63	−15.7	✓	79.4	90.5	94.3	96.9
Ozone Level Detection	class.	60	−13.6	✓	86.9	90.8	92.7	91.4
Website Phishing	class.	60	−11.1	✓	86.9	91.1	93.7	92.3
Maternal Health Risk	class.	70	−10.5	✓	86.3	90.9	94.5	95.5
Statlog (Shuttle)	class.	33	−4.8	✓	91.5	91.5	92.0	93.5
Mammographic Mass	class.	41	−4.2	✓	89.8	90.2	93.6	92.1
Congressional Voting Records	class.	26	−3.5	✓	84.4	88.3	93.2	90.3
CDC Diabetes Health Indicators	class.	13	−1.2	✓	89.3	89.9	90.5	90.1
Iranian Churn	class.	12	−1.2	✓	90.5	90.0	91.6	90.7
Sepsis Survival Minimal Clinical...	class.	100	−1.1	✓	88.9	99.2	100.0	–
Primary Tumor	class.	20	−0.8	✓	91.9	90.4	96.7	95.1
Haberman's Survival	class.	9	−0.7		89.2	89.6	94.9	93.6
Student Performance on an Entran...	class.	16	−0.7	✓	93.4	90.4	93.4	90.8
Differentiated Thyroid Cancer Re...	class.	6	−0.6		88.6	88.6	92.5	91.7
National Poll on Healthy Aging (...	class.	14	−0.5	✓	93.9	90.5	94.4	91.4
RT-IoT2022	class.	5	−0.5		94.1	94.1	94.1	–
Census-Income (KDD)	class.	2	−0.2		89.8	90.0	90.5	–
Liver Disorders	regre.	1	−0.2		88.8	90.1	94.1	92.5
Annealing	class.	2	−0.2		91.1	91.5	93.7	90.3
Soybean (Large)	class.	2	−0.1		90.9	90.9	95.9	91.7
Estimation of Obesity Levels Bas...	class.	1	−0.1		89.5	89.7	91.7	89.8

Table S4: Wave 2 (UCI repository), random forest, 20 splits. Columns as in Table S3

Data set	Task	$\bar{\pi}$	g^*	Flag	Std	Fix	Margin	Dist.
Online Shoppers Purchasing Inten...	class.	1	−0.1		90.0	90.1	90.9	90.2
Page Blocks Classification	class.	2	−0.1		90.5	90.1	91.1	90.7
Polish Companies Bankruptcy	class.	1	−0.1		90.1	90.1	90.7	–
Real Estate Valuation	regre.	6	0.0		91.1	91.5	94.5	93.3

S6 Benchmark results by family (H1)

Table S5: Benchmark data sets with repeats, 90% target. Gap: median new-case gap of standard calibration (pp) and worst data set; p: Holm-adjusted one-sided Wilcoxon p for H1; ρ: Spearman correlation of exposure and gap; Fix: median gap of separate calibration and lower bound of its 95% CI (pp); Size: median ratio of new-case set size, fix over standard

Task	Learner	Score	n	Gap	Worst	p	ρ	Fix	CI low	Size
class.	LR	lac	24	−0.5	−6.5	0.343	−0.58	+0.2	0.0	1.009
class.	LR	aps	24	−0.1	−1.5	1.000	−0.64	+0.2	−0.1	1.005
class.	RF	lac	24	−0.6	−6.5	0.420	−0.41	+0.1	−0.1	1.012
class.	RF	aps	24	+0.5	−1.2	1.000	−0.12	+0.1	−0.1	0.995
class.	LGBM	lac	24	−0.6	−6.7	0.092	−0.46	0.0	−0.2	1.011
class.	LGBM	aps	24	−0.2	−4.2	0.607	−0.84	+0.2	−0.2	1.003
class.	kNN	lac	24	+2.6	−2.8	1.000	−0.36	+3.6	+2.6	1.000
class.	kNN	aps	24	+0.1	−1.6	1.000	−0.11	+0.2	−0.1	1.001
class.	MLP	lac	24	−0.4	−6.9	0.395	−0.34	+0.2	−0.1	1.012
class.	MLP	aps	24	0.0	−4.9	1.000	−0.63	+0.3	0.0	1.005
regre.	LR	absres	12	−0.2	−16.8	1.000	−0.80	0.0	−0.5	1.026
regre.	RF	absres	12	−0.8	−15.8	0.881	−0.69	0.0	−0.2	1.043
regre.	LGBM	absres	12	−0.6	−16.9	1.000	−0.69	0.0	−0.1	1.027
regre.	kNN	absres	12	−0.2	−11.7	1.000	−0.37	0.0	−0.3	1.044
regre.	MLP	absres	12	−0.5	−10.8	0.833	−0.62	−0.1	−0.4	1.027
regre.	cqr	cqr	12	−0.6	−3.7	0.881	−0.55	0.0	−0.1	1.017

S7 Sampling-noise check

S8 Planted-leak test of the audit

S9 Validation of the tool on fresh files

Table S8: Post hoc. The installed *covercheck* command (defaults: random forest, 10 splits) on every wave-2 data set with at most 30,000 rows, downloaded afresh from UCI. New-case coverage (%) as tool / pipeline (pipeline: 20 splits, different preprocessing and seeds)

Data set	Rows	$\bar{\pi}$ (%)	Verdict	Std, tool/pipeline	Fix, tool/pipeline
Annealing	898	1.6	No Action Needed	90.9 / 91.1	91.2 / 91.5
Haberman's Survival	306	9.1	No Action Needed	90.1 / 89.2	90.3 / 89.6
Liver Disorders	345	1.4	No Action Needed	89.5 / 88.8	90.1 / 90.1

Table S8: Post hoc. The installed *covercheck* command (defaults: random forest, 10 splits) on every wave-2 data set with at most 30,000 rows, downloaded afresh from UCI. New-case coverage (%) as tool / pipeline (pipeline: 20 splits, different preprocessing and seeds)

Data set	Rows	$\bar{\pi}$ (%)	Verdict	Std, tool/pipeline	Fix, tool/pipeline
Page Blocks Classification	5,473	1.8	No Action Needed	90.8 / 90.5	90.3 / 90.1
Primary Tumor	339	20.4	Fix Advised	93.9 / 91.9	92.1 / 90.4
Soybean (Large)	307	1.6	No Action Needed	93.3 / 90.9	93.3 / 90.9
Congressional Voting Records	435	25.9	Fix Recommended	88.0 / 84.4	89.6 / 88.3
Mammographic Mass	961	40.8	Fix Recommended	89.8 / 89.8	90.8 / 90.2
Ozone Level Detection	5,070	60.1	Fix Recommended	87.8 / 86.9	91.6 / 90.8
Website Phishing	1,353	60.2	Fix Recommended	85.5 / 86.9	89.8 / 91.1
Online Shoppers Purchasing Inten…	12,330	1.2	No Action Needed	90.0 / 90.0	90.1 / 90.1
Real Estate Valuation	414	5.8	No Action Needed	90.8 / 91.1	90.7 / 91.5
Early Stage Diabetes Risk Predic…	520	62.6	Fix Recommended	79.6 / 79.4	88.4 / 90.5
Estimation of Obesity Levels Bas…	2,111	1.3	No Action Needed	89.3 / 89.5	89.6 / 89.7
Iranian Churn	3,150	12.4	Fix Advised	90.4 / 90.5	90.0 / 90.0
Student Performance on an Entran…	666	16.5	Fix Advised	94.3 / 93.4	90.2 / 90.4
NATICUSdroid (Android Permissions)	29,332	78.7	Fix Recommended	76.2 / 76.1	89.8 / 89.6
Maternal Health Risk	1,014	70.3	Fix Recommended	86.1 / 86.3	91.0 / 90.9
Differentiated Thyroid Cancer Re…	383	6.0	No Action Needed	86.9 / 88.6	87.2 / 88.6
National Poll on Healthy Aging (…	714	13.8	Fix Advised	94.8 / 93.9	91.2 / 90.5

S10 Software environment

Hardware: 4 logical CPUs (Intel(R) Xeon(R) Processor @ 2.10GHz), 16 GB memory. Software: Python 3.13.16; package versions are pinned in `code/requirements.txt` and recorded in `results/logs/environment.json`. One command, `python code/run_all.py`, reruns all experiments, the aggregation and the figures; `paper/build.sh` builds both documents.

References

[1] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani, L. Wasserman. Distribution-free predictive inference for regression. J Am Stat Assoc 113(523):1094–1111, 2018.

[2] V. Vovk. Conditional validity of inductive conformal predictors. Proc. Asian Conference on Machine Learning, PMLR 25:475–490, 2012.

[3] V. Vovk, A. Gammerman, G. Shafer. Algorithmic Learning in a Random World, 2nd edn. Springer, 2022.

Table S2: Selection of the two fresh collections: pool sizes, exclusions in the order applied, and the included data sets. "v1" denotes the benchmark and application data sets used earlier in the study. The full logs, with the reason for every data set, are in `results/reinforcement/prospective_selection*.json`

Step	Data sets
Wave 1: pool of 163 data sets	
predicted exposure < 0.01	76
id already used in v1	31
name already used in v1	16
size limits	11
download/parse failed	5
same data set already included under a lower id	4
included	**20**
Wave 2: pool of 209 data sets	
predicted exposure < 0.01	55
not exactly one target	40
size limits	38
size signature matches a v1 or wave-1 data set (conservative overlap rule)	24
name matches v1 or wave 1	20
fetch failed	3
non-numeric regression target	2
included	**27**

Table S6: Post hoc. Observed number of splits with new-case coverage below 87% against the number expected from pure sampling noise (Beta–binomial simulation given each split's numbers of calibration and new test rows), over all fresh data sets. p: probability under the noise model of at least the observed count

Method	Learner	Splits	Observed	Expected	Ratio	p
Standard	RF	939	184	87.3	2.11	<0.001
	LGBM	939	212	86.9	2.44	<0.001
	Linear	939	164	87.1	1.88	<0.001
Seen/new separately	RF	939	84	79.2	1.06	0.289
	LGBM	939	93	79.2	1.17	0.047
	Linear	939	83	79.4	1.05	0.336
+ safety margin	RF	939	16	15.1	1.06	0.446
	LGBM	939	25	14.9	1.67	0.006
	Linear	939	16	15.2	1.05	0.471

Table S7: Post hoc. A known share of test rows was copied into the training part of one random split of each of the first five duplicate-free benchmark data sets with at least 1,000 rows. Exact copies are all detected with no false positives; copies jittered by 0.1% of a feature's standard deviation are missed (kr-vs-kp has no numeric features, so its jittered copies remain exact)

Data set	Plant	Share (%)	Planted	Flagged	True pos.	False pos.	Recall (%)
kr-vs-kp	none	0	0	0	0	0	–
kr-vs-kp	exact	5	32	32	32	0	100
kr-vs-kp	jittered (0.1% SD)	5	32	32	32	0	100
kr-vs-kp	exact	20	128	128	128	0	100
kr-vs-kp	jittered (0.1% SD)	20	128	128	128	0	100
optdigits	none	0	0	0	0	0	–
optdigits	exact	5	56	56	56	0	100
optdigits	jittered (0.1% SD)	5	56	0	0	0	0
optdigits	exact	20	225	225	225	0	100
optdigits	jittered (0.1% SD)	20	225	0	0	0	0
credit-g	none	0	0	0	0	0	–
credit-g	exact	5	10	10	10	0	100
credit-g	jittered (0.1% SD)	5	10	0	0	0	0
credit-g	exact	20	40	40	40	0	100
credit-g	jittered (0.1% SD)	20	40	0	0	0	0
pendigits	none	0	0	0	0	0	–
pendigits	exact	5	110	110	110	0	100
pendigits	jittered (0.1% SD)	5	110	0	0	0	0
pendigits	exact	20	440	440	440	0	100
pendigits	jittered (0.1% SD)	20	440	0	0	0	0
electricity	none	0	0	0	0	0	–
electricity	exact	5	300	300	300	0	100
electricity	jittered (0.1% SD)	5	300	0	0	0	0
electricity	exact	20	1200	1200	1200	0	100
electricity	jittered (0.1% SD)	20	1200	0	0	0	0